

Algebra II with Trigonometry

Chapter 6.2

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

Solutions

1. Evaluate exactly:

$$\sin \frac{13\pi}{12} + \cos \frac{13\pi}{12}.$$

Note that

$$\frac{13\pi}{12} = \pi + \frac{\pi}{12}.$$

Hence

$$\sin \left(\pi + \frac{\pi}{12} \right) = -\sin \frac{\pi}{12}, \quad \cos \left(\pi + \frac{\pi}{12} \right) = -\cos \frac{\pi}{12}.$$

Therefore

$$\sin \frac{13\pi}{12} + \cos \frac{13\pi}{12} = - \left(\sin \frac{\pi}{12} + \cos \frac{\pi}{12} \right).$$

Now use

$$(\sin x + \cos x)^2 = \sin^2 x + \cos^2 x + 2 \sin x \cos x = 1 + \sin 2x.$$

With $x = \frac{\pi}{12}$,

$$\left(\sin \frac{\pi}{12} + \cos \frac{\pi}{12} \right)^2 = 1 + \sin \frac{\pi}{6} = 1 + \frac{1}{2} = \frac{3}{2}.$$

Since $\sin \frac{\pi}{12} + \cos \frac{\pi}{12} > 0$,

$$\sin \frac{\pi}{12} + \cos \frac{\pi}{12} = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}.$$

Thus

$$\sin \frac{13\pi}{12} + \cos \frac{13\pi}{12} = -\frac{\sqrt{6}}{2}.$$

$$\boxed{-\frac{\sqrt{6}}{2}}$$

2. A real number t satisfies

$$\sin t + \cos t = \frac{1}{3}.$$

Knowing t is in Quadrant II, find the exact value of $\tan t$.

Let

$$s = \sin t, \quad c = \cos t.$$

We are given

$$s + c = \frac{1}{3}.$$

Squaring,

$$s^2 + 2sc + c^2 = \frac{1}{9}.$$

Since $s^2 + c^2 = 1$,

$$1 + 2sc = \frac{1}{9} \quad \Rightarrow \quad 2sc = -\frac{8}{9} \quad \Rightarrow \quad sc = -\frac{4}{9}.$$

Now s and c are the roots of

$$u^2 - (s + c)u + sc = 0,$$

so

$$u^2 - \frac{1}{3}u - \frac{4}{9} = 0.$$

Multiply by 9:

$$9u^2 - 3u - 4 = 0.$$

Thus

$$u = \frac{3 \pm \sqrt{9 + 144}}{18} = \frac{3 \pm 3\sqrt{17}}{18} = \frac{1 \pm \sqrt{17}}{6}.$$

In Quadrant II, $\sin t > 0$ and $\cos t < 0$. Therefore

$$\sin t = \frac{1 + \sqrt{17}}{6}, \quad \cos t = \frac{1 - \sqrt{17}}{6}.$$

Hence

$$\tan t = \frac{\sin t}{\cos t} = \frac{1 + \sqrt{17}}{1 - \sqrt{17}}.$$

Rationalizing,

$$\tan t = \frac{(1 + \sqrt{17})^2}{1 - 17} = \frac{18 + 2\sqrt{17}}{-16} = -\frac{9 + \sqrt{17}}{8}.$$

$$\boxed{-\frac{9 + \sqrt{17}}{8}}$$

3. Without a calculator, determine the exact value of

$$\sec \frac{7\pi}{12} - \csc \frac{7\pi}{12}.$$

Write

$$\frac{7\pi}{12} = \frac{\pi}{3} + \frac{\pi}{4}.$$

Then

$$\cos \frac{7\pi}{12} = \cos \left(\frac{\pi}{3} + \frac{\pi}{4} \right) = \cos \frac{\pi}{3} \cos \frac{\pi}{4} - \sin \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} = \frac{\sqrt{2} - \sqrt{6}}{4}$$

and

$$\sin \frac{7\pi}{12} = \sin \left(\frac{\pi}{3} + \frac{\pi}{4} \right) = \sin \frac{\pi}{3} \cos \frac{\pi}{4} + \cos \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Therefore

$$\sec \frac{7\pi}{12} = \frac{4}{\sqrt{2} - \sqrt{6}}, \quad \csc \frac{7\pi}{12} = \frac{4}{\sqrt{6} + \sqrt{2}}.$$

So

$$\sec \frac{7\pi}{12} - \csc \frac{7\pi}{12} = \frac{4}{\sqrt{2} - \sqrt{6}} - \frac{4}{\sqrt{6} + \sqrt{2}}.$$

Rationalize each term:

$$\frac{4}{\sqrt{2} - \sqrt{6}} = \frac{4(\sqrt{2} + \sqrt{6})}{2 - 6} = -(\sqrt{2} + \sqrt{6}),$$

$$\frac{4}{\sqrt{6} + \sqrt{2}} = \frac{4(\sqrt{6} - \sqrt{2})}{6 - 2} = \sqrt{6} - \sqrt{2}.$$

Hence

$$\sec \frac{7\pi}{12} - \csc \frac{7\pi}{12} = -(\sqrt{2} + \sqrt{6}) - (\sqrt{6} - \sqrt{2}) = -2\sqrt{6}.$$

$$\boxed{-2\sqrt{6}}$$

4. If

$$\cos t = -\frac{5}{13} \quad \text{and} \quad \sin 2t < 0,$$

find the exact value of

$$\tan t + \cot t.$$

Since

$$\sin 2t = 2 \sin t \cos t < 0$$

and $\cos t = -\frac{5}{13} < 0$, it follows that $\sin t > 0$. Thus t is in Quadrant II.

Using $\sin^2 t + \cos^2 t = 1$,

$$\sin t = \sqrt{1 - \left(-\frac{5}{13}\right)^2} = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}.$$

Now

$$\tan t + \cot t = \frac{\sin t}{\cos t} + \frac{\cos t}{\sin t} = \frac{\sin^2 t + \cos^2 t}{\sin t \cos t} = \frac{1}{\sin t \cos t}.$$

Hence

$$\tan t + \cot t = \frac{1}{\left(\frac{12}{13}\right)\left(-\frac{5}{13}\right)} = \frac{1}{-\frac{60}{169}} = -\frac{169}{60}.$$

$$\boxed{-\frac{169}{60}}$$

5. Let

$$y(t) = 3 \cos(\pi t) + 4 \sin(\pi t).$$

Find the largest possible value of $y(t)$, and one value of $t \in [0, 2)$ at which it occurs.

Use the standard amplitude form:

$$a \cos x + b \sin x \leq \sqrt{a^2 + b^2}.$$

Here $a = 3$, $b = 4$, so

$$y(t) \leq \sqrt{3^2 + 4^2} = 5.$$

Thus the largest possible value is

$$\boxed{5}.$$

To find when equality occurs, write

$$3 \cos(\pi t) + 4 \sin(\pi t) = 5 \cos(\pi t - \phi),$$

where

$$\cos \phi = \frac{3}{5}, \quad \sin \phi = \frac{4}{5}.$$

Then the maximum 5 occurs when

$$\cos(\pi t - \phi) = 1,$$

i.e.

$$\pi t - \phi = 2\pi k.$$

Taking $k = 0$,

$$t = \frac{\phi}{\pi}.$$

Since $\phi = \arctan \frac{4}{3}$, one suitable value is

$$\boxed{t = \frac{1}{\pi} \arctan \frac{4}{3}}.$$

6. Find the exact value of

$$\left(\sin \frac{\pi}{12} \right) \left(\cos \frac{5\pi}{12} \right) + \left(\cos \frac{\pi}{12} \right) \left(\sin \frac{5\pi}{12} \right) - \tan \frac{3\pi}{4}.$$

Use the sine addition formula:

$$\sin A \cos B + \cos A \sin B = \sin(A + B).$$

With

$$A = \frac{\pi}{12}, \quad B = \frac{5\pi}{12},$$

we get

$$\left(\sin \frac{\pi}{12}\right) \left(\cos \frac{5\pi}{12}\right) + \left(\cos \frac{\pi}{12}\right) \left(\sin \frac{5\pi}{12}\right) = \sin \left(\frac{\pi}{12} + \frac{5\pi}{12}\right) = \sin \frac{\pi}{2} = 1$$

Also,

$$\tan \frac{3\pi}{4} = -1.$$

Therefore the whole expression equals

$$1 - (-1) = 2.$$

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